

MATH 2101 LINEAR ALGEBRA I
FALL SEMESTER 2025
TEST 2

Last Name (Family Name):

First Name (Given Name):

UID:

Instructions:

- (1) The question paper is in the **last page**. You may tear off the last page, but **return the questions paper** at the end of the test.
- (2) Write answers in the attached A4 papers. Writings in draft papers will NOT be graded.
- (3) Show your steps or reasonings clearly to receive full credits unless otherwise specified in the question.

Write your work in this page

Write your work in this page

Write your work in this page

Write your work in this page

Write your work in this page

Write your work in this page

Write your work in this page

Write your work in this page

Write your work in this page

Question Paper Test 2 (Fall 2025)
Writings on this page will not be graded

Name:

UID:

- (1) (3 points) Let $T : \mathbb{R}^7 \rightarrow \mathbb{R}^5$ be a linear transformation. Suppose $\text{rank}(T) = 3$. Find $\text{nullity}(T)$.

- (2) (5 points) Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \end{pmatrix}$$

Find a basis of the null space $N(A)$.

- (3) (3 points) Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Show that 5 is an eigenvalue of A .

- (4) (4 points) Prove or disprove: Let A be an $n \times n$ matrix. Then a scalar $\lambda \in \mathbb{R}$ is an eigenvalue of A if and only if λ is an eigenvalue of A^T .
- (5) (5 points) Let V be a vector space of dimension n . Let β be an ordered basis for V . Define $T : V \rightarrow \mathbb{R}^n$ given by $T(v) = [3v]_\beta$. Show that T is a linear transformation.
- (6) (5 points) Let V and W be two vector spaces. Suppose $\dim(V) < \dim(W)$. Show that there exists an injective linear transformation $T : V \rightarrow W$.